

Final Migration Report

Governed sandbox migration report

1. Executive Summary

Field	Value
Run ID	v2-6100d2bd-s4
Generated At	2026-06-25T10:40:48Z
Legacy Baseline	Spring Boot 2.1.6.RELEASE / Java 11
Current Application State	Spring Boot 4.0 / Java 21
Latest Completed Stage	Stage 4: springboot-3.5-java21-to-4.0-java21
Migration Duration	1722.412s
Approval Decision	approved
Transform Status	TRANSFORM_APPLIED_IN_SANDBOX
Build Status	BUILD_PASSED_IN_SANDBOX
Test Status	PASS_WITH_WARNINGS
Proof Level	compiled
Dependency Policy	PASS

This report describes the full migration journey from Spring Boot 2.1.6.RELEASE / Java 11 to Spring Boot 4.0 / Java 21. The migration was executed as a controlled sandbox modernization flow with human approval before transformation.

Migration completed for the full path Spring Boot 2.1.6.RELEASE / Java 11 -> Spring Boot 4.0 / Java 21 with approval decision approved, transform status TRANSFORM_APPLIED_IN_SANDBOX, build status BUILD_PASSED_IN_SANDBOX, and test status PASS_WITH_WARNINGS. Elapsed duration: 1722.412s. Primary change summary: Full migration path: Spring Boot 2.1.6.RELEASE / Java 11 -> Spring Boot 4.0 / Java 21

2. Migration Story

The application started from Spring Boot 2.1.6.RELEASE / Java 11 and progressed through the staged migration pipeline until it reached Spring Boot 4.0 / Java 21. The latest completed stage was Stage 4: springboot-3.5-java21-to-4.0-java21, where the final transition was Spring Boot 3.5 / Java 21 -> Spring Boot 4.0 / Java 21.

Migration flow followed during this run:

Analyze -> Plan -> Assess -> Human Approval -> Sandbox Transform -> Build Validation -> Test Validation -> Final Report

Migration Process

- * Legacy application baseline: Spring Boot 2.1.6.RELEASE / Java 11
- * Latest completed stage: Stage 4: springboot-3.5-java21-to-4.0-java21
- * Completed stage transition: Spring Boot 3.5 / Java 21 -> Spring Boot 4.0 / Java 21
- * Current application state: Spring Boot 4.0 / Java 21
- * Final executed target: Spring Boot 4.0 / Java 21
- * Human approval decision: approved
- * Sandbox transform result: TRANSFORM_APPLIED_IN_SANDBOX
- * Build result: BUILD_PASSED_IN_SANDBOX
- * Test result: PASS_WITH_WARNINGS
- * Proof level achieved: compiled
- * Repair loop outcome: NOT_IMPLEMENTED
- * Dependency policy outcome: PASS

3. Current Technical State

Area	Value
Legacy application baseline	Spring Boot 2.1.6.RELEASE / Java 11
Latest completed stage source	Spring Boot 3.5 / Java 21
Current application state	Spring Boot 4.0 / Java 21
Spring Framework target	7.0.x
Risk Level	high
Strategy	staged_openrewrite_sandbox
Fallback Profile	not captured
Production Allowed	false

4. Phase Status

Phase	Status	Explanation
Human Approval	approved	Human approval was required before sandbox transformation.
Sandbox Transform	TRANSFORM_APPLIED_IN_SANDBOX	Code changes were applied only inside the sandbox workspace.
Build Validation	BUILD_PASSED_IN_SANDBOX	The migrated sandbox candidate was validated by build execution.
Test Validation	PASS_WITH_WARNINGS	Existing tests were run against the migrated sandbox candidate.
Proof Level	compiled	This is the highest deterministic proof level reached during validation.
Repair Loop	NOT_IMPLEMENTED	Indicates whether repair logic was needed to stabilize the migration.
Dependency Policy	PASS	Shows whether dependency policy checks passed or raised review items.

5. Stage-By-Stage Journey

The migration was not a single jump. It was executed as a staged progression so each version boundary could be validated more safely.

Stage	Profile	Transition	Status	Duration
Stage 1	springboot-2.1.6-to-2.7-java11	Spring Boot 2.1.6.RELEASE / Java 11 -> Spring Boot 2.7 / Java 11	TRANSFORM_APPLIED_IN_SANDBOX	398.295s
Stage 2	springboot-2.7-to-3.5-java17	Spring Boot 2.7 / Java 11 -> Spring Boot 3.5 / Java 17	TRANSFORM_APPLIED_IN_SANDBOX	564.966s
Stage 3	springboot-3.5-java17-to-java21	Spring Boot 3.5 / Java 17 -> Spring Boot 3.5 / Java 21	TRANSFORM_APPLIED_IN_SANDBOX	253.857s
Stage 4	springboot-3.5-java21-to-4.0-java21	Spring Boot 3.5 / Java 21 -> Spring Boot 4.0 / Java 21	TRANSFORM_APPLIED_IN_SANDBOX	505.293s

Narrative highlights:

- * Stage 1 moved the application from Spring Boot 2.1.6.RELEASE / Java 11 to Spring Boot 2.7 / Java 11, with overall stage status TRANSFORM_APPLIED_IN_SANDBOX.
 - * Stage 2 moved the application from Spring Boot 2.7 / Java 11 to Spring Boot 3.5 / Java 17, with overall stage status TRANSFORM_APPLIED_IN_SANDBOX.
 - * Stage 3 moved the application from Spring Boot 3.5 / Java 17 to Spring Boot 3.5 / Java 21, with overall stage status TRANSFORM_APPLIED_IN_SANDBOX.
 - * Stage 4 moved the application from Spring Boot 3.5 / Java 21 to Spring Boot 4.0 / Java 21, with overall stage status TRANSFORM_APPLIED_IN_SANDBOX.
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6. What Changed

What Changed

The most important migration changes recorded for this run are listed below.

- * Full migration path: Spring Boot 2.1.6.RELEASE / Java 11 -> Spring Boot 4.0 / Java 21
 - * Java remained at 21.
 - * Spring Boot changed from 3.5 to 4.0.
 - * Spring Framework changed from not_captured to 7.0.x.
 - * Stage 1 (springboot-2.1.6-to-2.7-java11): Spring Boot 2.1.6.RELEASE / Java 11 -> Spring Boot 2.7 / Java 11
 - * Stage 2 (springboot-2.7-to-3.5-java17): Spring Boot 2.7 / Java 11 -> Spring Boot 3.5 / Java 17
 - * Stage 3 (springboot-3.5-java17-to-java21): Spring Boot 3.5 / Java 17 -> Spring Boot 3.5 / Java 21
 - * Stage 4 (springboot-3.5-java21-to-4.0-java21): Spring Boot 3.5 / Java 21 -> Spring Boot 4.0 / Java 21
 - * Executed OpenRewrite recipes: org.openrewrite.java.spring.boot4.UpgradeSpringBoot_4_0.
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7. Validation Outcome

Validated areas:

- * sandbox transform applied
- * Maven build passed

Not validated by this sandbox run:

- * SQL Server production behavior
- * production DB scripts
- * endpoint/business behavior
- * production secrets/JWT/keystore validity
- * deployment
- * PR creation/merge

This means the migration is well described technically, but any production-readiness conclusion still requires human review of runtime behavior, infrastructure compatibility, and environment-specific risks.

8. Spring Boot 4 Notes

These are the main points that deserve attention for the final Boot 4 state:

- * Spring Boot 4 target requires Spring Framework 7.x.
 - * Spring Boot 4 target carries Jakarta EE 11 / Servlet 6.1 baseline risk.
 - * Boot 3 deprecated APIs removed in Boot 4 must be reviewed.
 - * Spring Cloud compatibility must be reviewed before sandbox execution.
 - * Spring Security, Spring Data, Hibernate, and custom starter compatibility risk requires human review.
 - * javax.* leftovers must be treated as blockers for Boot 4 readiness.
 - * Maven version and Java runtime must match Boot 4 / Java 21 validation gates.
 - * Official Spring Boot guidance prefers upgrading to the latest 3.5.x before Boot 4; direct migration is sandbox-only and should fall back if unstable.
 - * OPENREWRITE_IMPACT_HIGH: OpenRewrite impact is high; manual review is required before execution.
 - * OPENREWRITE_HIGH_RISK_FILES: OpenRewrite reported high-risk files: 42.
 - * Spring Framework 7.x is required for Spring Boot 4.
 - * Jakarta EE 11 / Servlet 6.1 baseline applies.
 - * Spring Cloud compatibility must be reviewed.
 - * Spring Security, Spring Data, Hibernate, and custom starter risk requires human review.
 - * javax.* leftovers must be eliminated.
 - * Maven >= 3.6.3 and Java 21 runtime validation are required for this sandbox profile.
 - * Official Boot guidance prefers latest 3.5.x before Boot 4; direct migration must use the fallback profile if unstable.
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9. Repair And Stabilization

Field	Value
Enabled	false
Max Attempts	3
Attempts Used	0
Final Status	NOT_IMPLEMENTED
Ledger	
Safe Patch Applied	false
Human Review Required	false

11. Dependency Policy Review

Field	Value
Status	PASS
Risks	0
Blockers	0
Policy Patch Applied	false
Report	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\assessment\dependency_policy_report.json

12. Timing

Timing

Field	Value
Total duration	1722.412s
Timing report	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\performance\timing_report.json
Timing summary	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\performance\timing_summary.md

13. Related Artifacts

Artifact	Path
approval_decision	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\approval\approval_decision.json
approved_plan_lock	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\approval\approved_plan_lock.json
transformation_execution_plan	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\transformation\transformation_execution_plan.yaml
migration_ledger	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\workspaces\sandbox\migration\ledger.json
post_transform_test_report	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\test\post_transform\test_report.json
orchestration_summary	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\orchestration\orchestration_summary.json
timing_report	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\performance\timing_report.json
timing_summary	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\performance\timing_summary.md
final_migration_report	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\final\migration_report.json
final_migration_summary	C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\v2-6100d2bd-s4\final\migration_summary.md

14. Final Note

This document is intended to help a reviewer understand what happened during the migration, why the application is now in its current state, and which areas still require manual judgment. The deterministic run artifacts remain the source of truth.

POC-ready sandbox migration artifacts are captured under this run directory.